1. Purpose

Provide standardized guidelines for selecting engineering materials based on performance, cost, sustainability, and manufacturability.

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2. Selection Criteria

Materials shall be evaluated based on:

- Mechanical strength
- Weight
- Corrosion resistance
- Thermal conductivity
- Electrical conductivity
- Cost
- Availability
- Sustainability
- Manufacturability
- Recyclability

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3. Material Comparison

Material	Strength	Cost	Corrosion Resistance	Weight
Stainless Steel 316	High	High	Excellent	Heavy
Aluminum 6061	Medium	Medium	Good	Light
Titanium Alloy	Very High	Very High	Excellent	Medium
Carbon Fiber Composite	Very High	High	Excellent	Very Light
Polycarbonate	Low	Low	Good	Very Light

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4. Material Selection Process

1. Define functional requirements.

2. Identify environmental conditions.
3. Compare candidate materials.
4. Perform laboratory testing.
5. Evaluate manufacturing feasibility.
6. Conduct cost analysis.
7. Review sustainability impact.
8. Obtain engineering approval.

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5. Recommended Applications

Material	**Recommended Use**
Stainless Steel	Industrial frames
Aluminum	Sensor housings
Titanium	Aerospace components
Carbon Fiber	Lightweight structures
Polycarbonate	Protective covers

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6. Engineering Best Practices

- Select materials with proven industry performance.
- Consider lifecycle cost instead of only procurement cost.
- Validate material properties through laboratory testing.
- Ensure compliance with ISO, ASTM, and IEC standards.
- Maintain traceability for all approved material specifications.